

The Triadic Measurement Substrate

$$\bar{B} = 3\bar{\alpha}$$

Volume 1

Paper 2: Computational Dynamics of the Binary Alphabet: The Polarization Field, the Born-Rule Structural Correspondence, and the Emergence of Universal Computation

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Paper 2

Computational Dynamics of the Binary Alphabet: The Polarization Field, the Born-Rule Structural Correspondence, and the Emergence of Universal Computation

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Keywords: TMS, Binary Alphabet, Polarization Field, Born Rule, Majority Gate, Universal Computation, Programs, Stabilizer Dynamics, Computational Irreducibility

Status of Results

This paper distinguishes four categories of claim, each labeled where it appears in the text.

Derived. Results that follow from the five axioms, the Principle of Generative Economy, and the structures established in Paper 1 by explicit construction or proof. The polarization conjugate pair (ψ, Π_ψ) and the polarization wave equation $\partial^2 \psi / \partial t^2 - c^2 \nabla^2 \psi = 0$ (§ II), the propagation speed $c_{\text{TMS}} = 1/\sqrt{2}$ for ψ inherited from FCC geometry (§ 2.3), the three-way logical distinction among possibilities for synthetic-bit valuation and the selection of polarization-biased stochastic resolution (§ 1.2, § 3.3), the linear amplitude-to-probability mapping $P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi)/2$ from four geometric constraints (§ 4.1), the 3-input majority gate at every confirmation event (§ 5.1), and universal classical computation from {MAJ, constant} (§ 5.3) are derived in this sense. The PGE-Forced Wave-Form Theorem (§ VII): higher-order reinterpretation at any hierarchical boundary is wave-form by structural necessity, not by additional axiom. Used in Paper 5 § III to resolve the two-waves problem and in Paper 5 § IV for the QCA-to-QFT correspondence. The formal definition of a program as a configuration causally closed under T with $K(C) < |C|$ (§ 6.2) is a definition; the existence of programs is established by construction in Paper 3.

Structural correspondence. Results in which a TMS construction faithfully reproduces the skeleton of a known framework without claiming reproduction of every feature. The Born-rule correspondence of § 4.2 is the central case: the skeleton of “deterministic wave evolution between events plus amplitude-biased projective measurement at each event” is reproduced exactly; the linear mapping $(1 + \Pi_\psi)/2$ is the unique geometrically-forced form for two-outcome real-valued projective measurement. The full quadratic $|\cdot|^2$ mapping requires complex amplitudes and phase coherence, which are expected to arise from the macroscopic interference behavior of coupled (ψ, Π_ψ) propagation and are deferred. The label “structural correspondence” matches the [[4,2,2]] correspondence of Paper 1: skeleton reproduced, full formalism deferred.

Predictions / Conjectures. Falsifiable claims that follow from the framework but whose closed forms are not derived within this paper. The refinement $\beta = f(d, K_\psi)$ where K_ψ is the Kolmogorov complexity of the program running through the bit’s causal history (§ 6.4) is a sharpening of the Paper 1 conjecture, not a derivation of f . The empirical signatures of polarization refraction (§ 7.1) and the prediction that GUP corrections vary with a particle’s preparation history rather than being a fixed property of a species (§ 7.2) are

consequences of the framework's architecture; the explicit functional forms and falsification thresholds are deferred to Paper 3's refinements and Paper 5's simulation work.

Deferred derivations carried into Paper 3. Three results are stated in this paper and closed by derivation in Paper 3: the dissolution rate of non-T-closed configurations (§ 6.3, closed in Paper 3 § 2.5 as $\sqrt{2} \cdot L_R/\ell_p$ flops); the effective error-rate decay with coherence depth (§ 5.2, closed in Paper 3 § 2.5 as $(1/3)^d$ in the large-coherence limit); and the explicit conditions under which the stabilizer dynamics select program-bearing configurations against U_{sh} disruption (§ 6.3, closed in Paper 3 § 3.3 as the coherence bistability threshold). These are flagged here as deferrals rather than completed derivations.

This four-way labeling applies throughout the paper. Every claim in the body text falls into exactly one category.

Preface to Paper 2

This paper is the second installment in Volume 1 of the Triadic Measurement Substrate program. Volume 1, Paper 1 established the geometric and ontological architecture of the TMS: the agent manifold $M\alpha$, the triadic confirmation rule, the $1 \rightarrow 4$ multiplier, FCC necessity, the identification of the agent spacing with the Planck length, the conjugate field structure (φ, Π) , the $[[4,2,2]]$ structural correspondence, the binary alphabet $\bar{B} \in \{0, 1\}$ as the irreducible computational content of every confirmed bit, and the polarization field ψ as a parallel object propagating across the same FCC lattice. Paper 2 takes that architecture as given and develops its computational dynamics. No new axioms are introduced. No new free parameters are introduced. The Principle of Generative Economy continues to govern every selection.

Paper 2's central claim is that the same primitives that generate wave dynamics in Paper 1 — by geometric necessity — also generate biased stochastic measurement, universal computation, and stabilizer-protected programs. The Born rule, the majority gate, and computational irreducibility all emerge from the interaction of the polarization field ψ with the triadic confirmation rule and the maximum-entropy substrate U_{sh} . Computation in the TMS is not designed. It is the inevitable consequence of confirming a binary outcome against a polarization-modulated wave field.

Abstract. Building on the binary alphabet $\bar{B} \in \{0, 1\}$ and the parallel polarization field ψ established in Volume 1, Paper 1, we develop the computational dynamics of the TMS substrate. Three principal results are derived. First, the polarization field ψ obeys a homogeneous wave equation on the FCC lattice identical in form and propagation speed to that of the confirmation density ϕ , establishing ψ as a fully-fledged conjugate-pair dynamical object propagating at $c_{\text{TMS}} = 1/\sqrt{2}$ lattice spacings per flop. Second, at every confirmation event the resolution of a synthetic bit position into a binary value $\bar{B} \in \{0, 1\}$ is biased by the local incoming polarization wave amplitude against the unbiased stochastic kick from U_{sh} — yielding a linear amplitude-to-probability mapping $P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi)/2$ that constitutes a structural correspondence to the Born rule for projective measurement. Third, the geometric structure of triadic confirmation makes every confirmation event a probabilistic 3-input majority gate over the polarizations of its parent confirmations; the $[[4,2,2]]$ stabilizer dynamics from Paper 1 correct local errors and project the effective behavior toward deterministic majority logic at scale. Universal computation follows. We define a program as a configuration of the polarization landscape that reproduces itself under the flop operator T while remaining algorithmically compressible ($K(C) < |C|$), and show that the stabilizer dynamics provide an automatic selection mechanism: configurations that satisfy stabilizer parity and reproduce under T persist; configurations that do not dissolve into U_{sh} — like noise. The $\beta(d, K_\psi)$ prediction sharpened in Paper 1 acquires its computational meaning here: the depth and polarization-complexity of a bit’s causal history is the depth and complexity of the program running through it. As in Paper 1, no free numerical parameters are introduced; every result follows by geometric and informational necessity from the axioms.

Paper 2 engages directly with two active programs deriving quantum mechanics from substrate dynamics: ’t Hooft’s Cellular Automaton Interpretation, which derives apparent quantum stochasticity from a deterministic underlying automaton observed in the wrong basis (’t Hooft, 2016), and Gorard’s derivation of quantum measurement from causal invariance and multiway-graph collapse in the Wolfram model (Gorard, 2020). The distinctive move of TMS relative to these programs is the polarization-biased resolution rule on a single confirmed lattice, which derives the linear amplitude-to-probability mapping from signal conservation and the maximum-entropy character of U_{sh} without invoking either superposition over multiway branches or a hidden deterministic preferred basis.

I. Foundations from Paper 1 and the Computational Program

1.1 Inherited Structure

Paper 1 establishes the following structures, which Paper 2 takes as given without re-derivation:

- The agent manifold $M\alpha$ with agent spacing $\ell_0 \equiv \ell_p$ (Axiom 2; Section 3.2 of Paper 1).
- The FCC causal stack with face-diagonal nearest-neighbor connectivity (Section 2.4 of Paper 1).
- The triadic actualization rule $\bar{B} = 3\bar{\alpha}_2$ in 2D and $\bar{B} = 4\bar{\alpha}_3$ in 3D (Sections 2.1, 2.5 of Paper 1).
- The $1 \rightarrow 4$ multiplier with informational closure (Section 4 of Paper 1).
- The binary alphabet $\bar{B} \in \{0, 1\}$ with polarization $\chi \equiv 2\bar{B} - 1 \in \{-1, +1\}$ (Section 1.4 of Paper 1).
- Polarization-carrying touch vector signals (Section 1.3 of Paper 1).
- The flop as the atomic unit of state change with no sub-flop time (Section 1.5 of Paper 1).
- The conjugate field structure (ϕ, Π) and the wave equation for ϕ at $c_{\text{TMS}} = 1/\sqrt{2}$ (Section 6 of Paper 1).
- The $[[4,2,2]]$ structural correspondence with logical qubit LQ1 = bit value, LQ2 = causal-history inheritance, jointly protected by S_x and S_z (Section 5 of Paper 1).
- The polarization field ψ as parallel propagating object on the same FCC lattice (Section 6.8 of Paper 1).
- U_{sh} as the maximum-entropy substrate from which both latent bits and stochastic disruptions arise, with no preferred direction, scale, or target (Axiom 0; Section 6.6 of Paper 1).

These structures are the entirety of the foundation. Paper 2 introduces no further primitives.

Remark 0.0.1 (Forward-Compatibility with the Sublattice Bifurcation). At every hierarchical level $k \geq 1$, the meta-FCC splits into two interpenetrating sublattices A and B running independent $[[4,2,2]]$ dynamics (Paper 1 § 2.4 corollary, full development in Paper 5 § 6.1.3 and § 5.3). This affects no Paper 2 derivations at the substrate level but is relevant when the polarization-field and Born-rule arguments are recursively applied at meta-levels.

1.2 The Computational Question

Paper 1 closed by establishing that every confirmed bit carries a binary value. It deferred the question of how that value is determined at the moment of confirmation. Paper 2 answers it.

The question is sharper than it first appears. The synthetic bit positions of the next layer S_{n+1} are geometrically fixed by the $1 \rightarrow 4$ multiplier acting on the parent confirmations in S_n . Every synthetic bit position is a geometric appointment — a location where a confirmation event must occur at the next flop. The question is not whether the appointment is kept (informational closure forces it to be) but with what value.

Three logically distinct possibilities exist:

- **Inherited value.** The synthetic bit takes a deterministic function of its parents' values. This requires specifying that function — a rule mapping $(\chi_1, \chi_2, \chi_3) \rightarrow \chi_{\text{child}}$ — which is an external program imposed on the substrate. It violates the Principle of Generative Economy.
- **Independent randomness.** The synthetic bit takes an unbiased random value from U_{sh} , with no influence from the parents. This violates Axiom 3 (signal conservation): the polarizations carried by the 12 outgoing signals from the parent confirmations are then dissipated rather than expressed in the next layer's confirmations.
- **Polarization-biased stochastic resolution.** The synthetic bit's resolution is stochastic, but the probability distribution is biased by the local incoming polarization carried by its parents' touch vector signals. The bias is geometric. The randomness is the residual contribution from U_{sh} .

Possibility 3 is the unique selection under the joint constraints of Axiom 3 and the Principle of Generative Economy. Polarization signals are conserved by being expressed as bias, not dissipated; no external rule is imposed; U_{sh} provides the stochastic component as it does at every scale of the TMS. Sections III and IV formalize this resolution mechanism and identify it as a structural correspondence to the Born rule for projective measurement.

II. The Polarization Field ψ and its Wave Equation

2.1 Definition of the Polarization Field

Definition 0.0.2 (Polarization Field). Let $\psi(\bar{r}, \tau)$ denote the local coarse-grained polarization — the scalar field representing the mean polarization of confirmed bits at any point in the FCC lattice:

$$\psi(\bar{r}, \tau) = \langle \chi \rangle(\bar{r}, \tau) = \langle 2\bar{B} - 1 \rangle(\bar{r}, \tau)$$

averaged over a region containing many lattice sites but small compared to wave scales. ψ ranges over $[-1, +1]$. $\psi = +1$ corresponds to a region of pure confirmed-presence; $\psi = -1$ corresponds to a region of

pure confirmed-absence; $\psi = 0$ corresponds to a region of balanced binary content. ψ is defined wherever the confirmation density ϕ is non-zero; in regions of vanishing ϕ , ψ is undefined.

The polarization field is distinct from the confirmation field. ϕ counts whether interstices are confirmed; ψ records the binary content of those confirmations. The two fields are independent in the bulk: any value of $\psi \in [-1, +1]$ is compatible with any value of $\phi \in (0, 1]$.

2.2 The Conjugate Excitation Π_ψ

By the same argument as Section 6.3 of Paper 1, the FCC cascade applied to the polarization field generates a forward-propagating excitation that must be tracked separately from the field itself. Define:

$$\Pi_\psi(S_n) = \text{the inherited polarization pressure being driven forward into } S_{n+1}$$

ψ records the polarization that has locked in past layers; Π_ψ records the polarization that the touch vector network is carrying forward to the next layer. The pair (ψ, Π_ψ) is the polarization analogue of the (ϕ, Π) pair from Paper 1.

Lemma 0.0.3 (Polarization Conjugacy). *The polarization field ψ and the forward-propagating polarization excitation Π_ψ constitute a co-generated conjugate pair, arising necessarily from the interaction between the polarization-carrying $1 \rightarrow 4$ cascade and U_{sh} disruption.*

Proof. The argument of Paper 1 Lemma (Field Conjugacy) applies in full, with three substitutions: (a) confirmation events become polarized confirmation events; (b) signals become polarized signals; (c) the conserved quantity is not the count of signals but the signed sum of their polarizations. Step 1 (2D first-order closure), Step 2 ($1 \rightarrow 4$ cascade), Step 3 (U_{sh} regulation produces dynamic stability), and Step 4 (signal conservation forces conjugate coupling) carry through unchanged. By signal conservation applied to polarization content:

$$\partial \psi / \partial \tau = \Pi_\psi$$

$$\partial \Pi_\psi / \partial \tau = c^2 \Delta_{\text{FCC}} \psi$$

Step 5 (elimination) yields:

$$\partial^2 \psi / \partial \tau^2 = c^2 \Delta_{\text{FCC}} \psi$$

The conjugate-pair structure is not an analogy. It is the same theorem applied to a different conserved quantity.

□

2.3 The Wave Equation for ψ

Theorem 0.0.4 (Polarization Wave Equation). *The polarization field ψ satisfies the discrete wave equation on the FCC lattice:*

$$\partial^2 \psi_i / \partial \tau^2 = c_{\text{TMS}}^2 \Delta_{\text{FCC}} \psi_i$$

with propagation speed $c_{\text{TMS}} = 1/\sqrt{2}$ lattice spacings per flop. In the continuum limit:

$$\partial^2 \psi / \partial t^2 - c^2 \nabla^2 \psi = 0$$

Proof. The propagation-speed argument of Paper 1 Section 6.5 depends only on the FCC lattice geometry — specifically, on the second-order tensor $\Sigma_j r_j \mu r_j \nu = 4\ell_p^2 \delta \mu \nu$ computed over the 12 nearest neighbors. This tensor is a property of the lattice, not of the field being propagated. The same argument applies to any scalar field that propagates by nearest-neighbor coupling on the FCC lattice. Therefore $c_{\text{TMS}} = 1/\sqrt{2}$ for ψ . The continuum limit follows by the Laplacian convergence argument of Paper 1 Section 6.6, again depending only on lattice geometry. □

Corollary 0.0.5 (No New Parameter). *The polarization field introduces no new propagation speed, no new coupling constant, and no new length scale. It inherits ℓ_p , t_p , and c from Paper 1 without modification.*

2.4 Independence of φ and ψ in the Bulk

The confirmation field and the polarization field share a substrate but propagate independently in the bulk. A wave in φ describes how confirmation density propagates regardless of binary content; a wave in ψ describes how polarization content propagates regardless of confirmation density. The two fields couple only at confirmation events themselves — the moments at which a new bit is added to the causal record. The coupling mechanism is the subject of Section III.

III. Confirmation Events: The Coupling of φ and ψ

3.1 The Atomicity of the Flop Revisited

Paper 1 Section 1.5 established that a flop is indivisible: the locking of a bit, the firing of 12 outgoing signals, the consumption of those signals as 4 synthetic bit positions, and the Z-displacement to the next layer all occur as a single act with no sub-flop time. Paper 2 examines the substructure of this single act with respect to the polarization landscape — not by subdividing the flop temporally (which is forbidden) but by identifying the geometric components present at the instant the flop occurs.

At the moment a confirmation event occurs at synthetic bit position x in layer S_{n+1} :

- Three touch vector signals are arriving at x , one from each of the three confirming agents in the parent layer S_n . Each signal carries the polarization of its source confirmation: $\chi_1, \chi_2, \chi_3 \in \{-1, +1\}$.
- U_{sh} contributes its standard unbiased stochastic input — the same maximum-entropy field that supplies latent bits across the substrate (Axiom 0; Paper 1 Section 1.4).
- The triadic agreement rule fires, locking the synthetic bit position into a confirmed state and selecting its binary value $\bar{B} \in \{0, 1\}$.

These components are not temporally sequential — they are the geometric content of the single indivisible act.

3.2 The Local Polarization Amplitude

Definition 0.0.6 (Local Polarization Amplitude). At a synthetic bit position x in layer S_{n+1} , the local polarization amplitude is the average of the polarizations carried by the three incoming touch vector signals from x 's parent agents in S_n :

$$\Pi_\psi(x) = (\chi_1 + \chi_2 + \chi_3)/3$$

$\Pi_\psi(x)$ ranges over the four discrete values $\{-1, -1/3, +1/3, +1\}$, corresponding to the four possible parent configurations under the symmetry of the three input slots.

$\Pi_\psi(x)$ is the same Π_ψ that propagates as the conjugate excitation of ψ in Section II — measured at the specific synthetic-bit position x . The bulk wave equation describes how Π_ψ evolves across the lattice between confirmation events; the local polarization amplitude at x is its value at the moment a confirmation occurs there.

3.3 Synthetic Bits Carry Position, Not Value

Proposition 0.0.7. *A synthetic bit, prior to its confirmation event, is a geometric appointment — a position in S_{n+1} at which a confirmation must occur — and does not carry a binary value of its own. Its value is determined at the moment of confirmation by the joint action of incoming polarization and U_{sh} .*

Proof. By the $1 \rightarrow 4$ multiplier (Paper 1 Section 4.2), each primary confirmation produces 4 synthetic bit positions through the geometric consumption of its 12 outgoing signals. The geometric act that produces a synthetic bit position is the projection of touch vectors forward through the FCC lattice; this projection determines location but does not by itself determine binary value. Conferring a value on the synthetic bit at the moment of its production would require selecting between Possibility 1 (inherited value rule, violating Generative Economy) and Possibility 2 (independent randomness at production, violating Axiom 3) from Section 1.2. Possibility 3 — biased stochastic resolution at the moment of confirmation — is the unique geometrically consistent option.

□

This distinction between geometric appointment and valued outcome resolves an apparent ambiguity in the substrate's dynamics: synthetic bits and confirmed bits are the same physical event seen at production-time and stabilization-time respectively. The $1 \rightarrow 4$ multiplier produces appointments; the confirmation event produces values.

IV. The Born-Rule Structural Correspondence

4.1 The Resolution Probability

Theorem 0.0.8 (Polarization-Biased Resolution). *Given a synthetic bit position x with local polarization amplitude $\Pi_\psi(x) \in [-1, +1]$, the probability that the confirmation event at x resolves as $\bar{B} = 1$ is:*

$$P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi) / 2$$

and equivalently the probability of $\bar{B} = 0$ is:

$$P(\bar{B} = 0 \mid \Pi_\psi) = (1 - \Pi_\psi) / 2$$

These probabilities sum to unity by construction and are non-negative for all $\Pi_\psi \in [-1, +1]$.

Proof. The resolution probability must satisfy four constraints, each forced by a primitive of the TMS:

- $\Pi_\psi = +1$ (all parents confirmed $\bar{B} = 1$) $\rightarrow P(\bar{B} = 1) = 1$. Required by Axiom 3: if all three incoming signals carry $\chi = +1$, dissipating any of that polarization through a non-certain resolution would violate signal conservation in the polarization content.
- $\Pi_\psi = -1$ (all parents confirmed $\bar{B} = 0$) $\rightarrow P(\bar{B} = 1) = 0$. Required by the same argument with the opposite polarization.
- $\Pi_\psi = 0$ (no net bias) $\rightarrow P(\bar{B} = 1) = 1/2$. Required by the maximum-entropy character of U_{sh} (Axiom 0): in the absence of bias, the stochastic kick from U_{sh} resolves the synthetic bit with equal probability between its two possible values.
- Monotonicity and continuity in Π_ψ . Required by the Principle of Generative Economy: any non-monotonic or discontinuous dependence on Π_ψ would introduce structure beyond what is forced by (i)–(iii), violating the minimum-sufficient selection criterion.

The unique function satisfying (i), (ii), (iii), and (iv) is the linear map $P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi)/2$. No alternative mapping satisfies all four constraints without an additional parameter (such as a non-linear curve shape).

□

4.2 Connection to the Born Rule

The resolution probability $P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi)/2$ is structurally analogous to the Born rule for projective measurement in quantum mechanics (Born, 1926). The correspondence consists of three parallel components:

- Unitary propagation between measurements $\leftrightarrow$ wave propagation between confirmation events. In QM, the state evolves deterministically via the Schrödinger equation between measurements. In TMS, the polarization field ψ and its excitation Π_ψ evolve deterministically via the FCC wave equation between confirmation events.
- Measurement outcome biased by amplitude $\leftrightarrow$ resolution outcome biased by Π_ψ . In QM, the outcome of a projective measurement is stochastic but its probability is determined by the local wave amplitude. In TMS, the outcome of a confirmation event is stochastic but its probability is determined by the local polarization amplitude.
- Born probability rule $\leftrightarrow$ linear polarization-to-probability mapping. In QM, $P(\text{outcome}) = |\langle \text{outcome} \mid \psi \rangle|^2$ — a quadratic mapping from complex amplitude to probability. In TMS, $P(\bar{B} = 1) = (1 + \Pi_\psi)/2$ — a linear mapping from real polarization to probability.

Status of the Correspondence. The mapping $(1 + \Pi_\psi)/2$ is the Born rule restricted to two-outcome projective measurement on real-valued amplitudes with no phase information. The recovery of the full quadratic $|\cdot|^2$ mapping requires complex amplitudes and phase coherence — structures that the TMS does not yet exhibit at the level of the polarization field ψ alone. Phase information in the TMS is carried by the time-evolution of the (ψ, Π_ψ) pair across many flops, not by an instantaneous complex amplitude at any single lattice site. The derivation of the full complex Born rule from the macroscopic interference behavior of the polarization wave equation is reserved for subsequent work. The present claim is one of structural correspondence — the same skeleton of “deterministic wave evolution + amplitude-biased

projective measurement” — exactly parallel in status to the $[[4,2,2]]$ structural correspondence of Paper 1, where the stabilizer algebra is faithfully reproduced but the full quantum amplitude formalism is deferred. $\square$

Remark 0.0.9. This identification dissolves the standard interpretive distinction between unitary evolution and measurement collapse. In the TMS, both are aspects of the same substrate dynamics. The wave equation governs propagation; the confirmation event resolves the value; U_{sh} supplies the irreducible stochasticity. No additional postulate is required to separate the two regimes. The “collapse” of standard quantum mechanics is, in TMS terms, the per-flop projection of polarization amplitude onto the binary alphabet.

4.3 Positioning Against Parallel Programs

Several active programs aim to derive quantum measurement from substrate-level dynamics, and the polarization-biased resolution rule of § 4.1 should be situated explicitly against the two most directly comparable proposals.

’t Hooft’s Cellular Automaton Interpretation of quantum mechanics (’t Hooft, 2016) proposes that the apparent randomness of quantum measurement is the consequence of observing a deterministic underlying cellular automaton in the “wrong basis” — a basis in which the deterministic dynamics appear as superposition and the deterministic resolution appears as stochastic collapse. The CAI’s deterministic ontology is realized in TMS in a partial sense: the polarization-biased resolution rule fixes the probability of each outcome from the local polarization amplitude, so the substrate is deterministic in its bias but stochastic in its residual selection. The residual stochasticity comes from U_{sh} , which by Axiom 0 is algorithmically incompressible — not a hidden deterministic dynamic but an irreducible source. TMS is therefore neither a hidden-variable theory in the strict CAI sense nor a Copenhagen-style ontological collapse theory; it is a third option in which the bias is geometric and the residual randomness is forced by the maximum-entropy substrate.

Gorard’s derivation of quantum measurement in the Wolfram model (Gorard, 2020) proceeds through the multiway evolution graph: a single substrate state evolves into many possible successors under non-confluent hypergraph rewriting, and observers impose effective causal invariance via a Knuth-Bendix completion that collapses distinct multiway branches into a single observed thread. TMS achieves a similar outcome through a different mechanism. There is no multiway evolution graph — every flop produces a single confirmed configuration on a single FCC lattice, determined by the polarization-biased resolution rule with U_{sh} supplying the stochastic component. The substrate of TMS is not a superposition of histories projected onto a single observed branch; it is a single history whose per-event outcomes are stochastic by the maximum-entropy properties of U_{sh} . The two pictures converge on the operational appearance of quantum measurement — wave-like evolution between events, probabilistic resolution at each event — while disagreeing on whether the substrate carries one history or many.

The decisive structural difference common to both contrasts is that TMS derives the linear amplitude-to-probability mapping $(1 + \Pi_{\psi})/2$ from signal conservation and the maximum-entropy character of U_{sh} alone, without invoking either a hidden deterministic basis (’t Hooft) or a multiway branching structure (Gorard). The mapping is the unique function satisfying the four geometric constraints of § 4.1, and its emergence requires no additional ontological machinery beyond what was already established in Paper 1.

V. Universal Computation from Triadic Confirmation

5.1 The 3-Input Majority Gate

Theorem 0.0.10 (Majority Gate Emergence). *Every confirmation event in the TMS implements a probabilistic 3-input majority gate over the polarizations of its parent confirmations, with the bias toward the deterministic majority outcome set by the magnitude of the polarization amplitude $|\Pi_\psi|$.*

Proof. From Section 3.2, the local polarization amplitude at synthetic bit position x is the average of three parent polarizations: $\Pi_\psi(x) = (\chi_1 + \chi_2 + \chi_3)/3$. The four possible parent configurations (up to permutation) and their resolution probabilities are:

- $(+1, +1, +1)$: $\Pi_\psi = +1$, $P(\bar{B} = 1) = 1$, $P(\bar{B} = 0) = 0$.
- $(+1, +1, -1)$: $\Pi_\psi = +1/3$, $P(\bar{B} = 1) = 2/3$, $P(\bar{B} = 0) = 1/3$.
- $(+1, -1, -1)$: $\Pi_\psi = -1/3$, $P(\bar{B} = 1) = 1/3$, $P(\bar{B} = 0) = 2/3$.
- $(-1, -1, -1)$: $\Pi_\psi = -1$, $P(\bar{B} = 1) = 0$, $P(\bar{B} = 0) = 1$.

In the deterministic limit $|\Pi_\psi| \rightarrow 1$ (achieved when all three parents agree), the resolution is certain to match the parents. In the intermediate cases, the resolution is biased toward the majority value with probability $2/3$. This is the canonical probabilistic majority function on three binary inputs.

□

5.2 Stabilizer Repair and Effective Determinism

The $1/3$ probability of “minority” resolutions in the intermediate cases is not noise to be ignored. It is a single-bit error rate. By Paper 1 Section 5, every tetrahedral confirmation unit detects and repairs single-bit errors within the same flop via the $[[4,2,2]]$ stabilizer dynamics. The synthetic bits produced at the current flop repair the missing or anomalous confirmation, restoring stabilizer parity.

Theorem 0.0.11 (Effective Determinism at Scale). *When the $[[4,2,2]]$ stabilizer repair mechanism is applied across a connected region of the lattice in which polarizations are coherent, the effective behavior of the substrate at scale is deterministic 3-input majority computation.*

Proof sketch. The single-bit error rate per confirmation in the intermediate cases is $1/3$. The $[[4,2,2]]$ code has distance 2 — any single-bit error is detected, and the synthetic repair process applied at the next flop projects the local configuration toward the nearest stabilizer-satisfying state. In a coherent region (where the polarization landscape varies slowly on the scale of single confirmation events), the nearest stabilizer-satisfying state is the deterministic majority result. Therefore the effective per-confirmation error rate after stabilizer projection decays geometrically with depth into the coherent region. At causal depth d , the effective probability of an erroneous resolution is bounded above by $(1/3)^d$, which vanishes rapidly. The substrate’s macroscopic behavior is therefore effectively deterministic majority computation in any coherent region. A complete derivation of the decay rate and the exact dependence on coherence-region geometry is provided in Paper 3 § 2.5.

□

5.3 Universality

Corollary 0.0.12 (Computational Universality). *The TMS substrate supports universal classical computation.*

Proof. The 3-input majority gate is universal in combination with a constant input. Specifically, $\text{MAJ}(x, y, 0) = \text{AND}(x, y)$, $\text{MAJ}(x, y, 1) = \text{OR}(x, y)$, and NOT can be implemented by polarization inversion — a

physical reflection of the polarization landscape across the $\chi = 0$ axis, achievable in TMS by routing through a confirmation region whose parent polarizations are uniformly opposite to the input. The set {AND, OR, NOT} is functionally complete (Shannon, 1948).

□

The substrate is therefore a universal computer by geometric necessity. No additional design is required — the triadic confirmation rule, applied to a binary alphabet on a fault-tolerant FCC lattice, is sufficient.

VI. Programs, Selection, and Computational Irreducibility

6.1 The Flop Operator and Configurations

Definition 0.0.13 (Flop Operator). The flop operator T maps the polarization landscape at layer S_n to the (probabilistic) polarization landscape at layer S_{n+1} :

$$T : \psi(S_n) \mapsto \psi(S_{n+1})$$

T is defined by the local resolution rule of Section IV combined with the global geometric projection of the $1 \rightarrow 4$ multiplier from S_n into S_{n+1} . T is probabilistic in general (due to the U_{sh} contribution) and becomes effectively deterministic in coherent regions (per Section 5.2).

Definition 0.0.14 (Configuration). A configuration C is a finite spatial pattern of confirmed polarizations across some bounded region of the lattice extending across a causal depth window d :

$$C = \{\chi(x, \tau) : x \in R, \tau \in [\tau_0, \tau_0 + d]\}$$

where R is the spatial support of the configuration.

6.2 Programs

Definition 0.0.15 (Program). A configuration C is a program of period n iff:

- Causal closure under T : $T^n[C] = C$ with probability approaching 1 in the limit of large coherent surround. The configuration's polarization landscape reproduces itself, possibly up to lattice translation, after n flops.
- Algorithmic compressibility: $K(C) < |C|$. There exists a description of C shorter than its raw enumeration.

Programs of period 1 are static patterns. Programs of period $n > 1$ are oscillators or propagating patterns. A program of period n that translates by some lattice vector v under T^n is a propagating program — the TMS analogue of a particle, though the connection to physical matter is deferred to subsequent work.

Remark 0.0.16. Condition (b) is the formal criterion that distinguishes a program from a coincidence in U_{sh} . Random configurations drawn directly from the maximum-entropy substrate satisfy $K(C) \approx |C|$ almost surely (Kolmogorov, 1965; Axiom 0). Only configurations with $K(C) < |C|$ have structure beyond noise — beyond what U_{sh} would produce by accident. Programs in the TMS are not configurations that “look organized” by inspection; they are configurations that are organized in the strict Kolmogorov sense.

Remark 0.0.17 (Status of $K(C)$ in the Program Definition). The Kolmogorov complexity $K(C)$ is not computable in general (Kolmogorov, 1965). The definition of a program via $K(C) < |C|$ therefore raises an

apparent question: how does the substrate select program-bearing configurations if it cannot compute the criterion that defines them?

The answer is that the substrate does not compute $K(C)$. $K(C)$ appears here as a property of configurations selected by the substrate's geometric dynamics, not as a criterion the substrate evaluates and applies. The selection mechanism is the stabilizer dynamics plus T-closure plus the coherence threshold — all of which are local geometric operations on the polarization landscape, computable at every flop by the substrate's own update rule. Configurations that survive this selection happen to satisfy $K(C) < |C|$ as a consequence of two structural facts: (a) random samples drawn directly from U_{sh} have $K(C) \approx |C|$ almost surely by Kolmogorov's incompressibility theorem; (b) configurations that survive geometric selection are by definition not random samples from U_{sh} , since they exhibit the structural regularities required by T-closure and stabilizer parity.

The implication runs in one direction only. Geometric selection implies $K(C) < |C|$ as an emergent property of the survivors. $K(C) < |C|$ does not need to be checked, computed, or enforced — it is what the selected configurations turn out to have. The substrate operates entirely on local geometric criteria; the algorithmic-compressibility characterization is the external observer's description of what those criteria pick out.

This distinction matters for the exhaustion analysis of Paper 4 § VI. Exhaustion is not a substrate event triggered by the substrate “noticing” that novelty has run out; it is a property of the configuration space that emerges from the finite size of $\Sigma(R)$. The substrate continues operating throughout. The transition to reinterpretation at $\tau > \tau^*(R)$ is forced by the structural situation (Axiom 4 forbids overwriting the confirmed bits; Axiom 3 forbids losing the signals that produced them; direct sampling extracts no further novelty from a saturated configuration space) — not by the substrate computing anything about $K(C)$.

6.3 The Selection Mechanism

The TMS provides an automatic selection mechanism that separates programs from noise without any external evaluator.

Theorem 0.0.18 (Stabilizer Selection). *Configurations that satisfy $[[4,2,2]]$ stabilizer parity and are causally closed under T persist across flops with probability approaching 1 in the limit of large coherent surround. Configurations that violate stabilizer parity are repaired toward the nearest parity-satisfying configuration. Configurations that are not causally closed dissolve into U_{sh} — like noise across $O(d)$ flops, where d is the causal depth across which they would need to maintain coherence.*

Proof sketch. By Paper 1 Section 5, single-bit deviations from any locally stabilizer-satisfying configuration are detected and repaired within the same flop. Configurations that satisfy stabilizer parity are local fixed points of the repair process. Configurations that violate parity are projected by repair toward the nearest stabilizer-satisfying state. For a configuration to persist as a program, it must additionally close under T — i.e., the natural forward evolution of its polarization landscape must reproduce its current state. Configurations that satisfy stabilizer parity but not causal closure under T are repaired in place but drift under wave dynamics, dissolving across $O(d)$ flops where d is the spatial extent of the configuration relative to the wave propagation speed $c_{TMS} = 1/\sqrt{2}$. A complete derivation of the dissolution rate is provided in Paper 3 § 2.5. $\square$

Remark 0.0.19. This is selection without natural selection. The TMS substrate distinguishes programs from noise by the same mechanism it distinguishes confirmed states from unconfirmed states: stabilizer projection. There is no separate evaluator, no fitness function, no environmental judge. The geometry is the selector. Configurations that compute persist; configurations that do not compute dissolve.

6.4 Computational Irreducibility of Programs

The arc from Axiom 0 to programs is a continuation of the arc that Paper 1 traced from Axiom 0 to the GUP. A program of causal depth d is a configuration whose existence requires d flops of confirmation to be specified. By Axiom 0, the U_{sh} substrate from which each layer's contribution to the program is drawn is algorithmically incompressible: $K(n) \approx n$. The down-vector chain encoding a program of depth d therefore has a lower bound on its compressibility set by the U_{sh} contributions at each layer.

For programs (configurations with $K(C) < |C|$), this lower bound is non-trivial but non-vanishing: programs compress because of their internal structure, but they do not compress arbitrarily because their causal history records irreducible contributions from U_{sh} at every layer.

Proposition 0.0.20 (Program Irreducibility). *A program of causal depth d has minimum description length bounded below by $K_\psi(d)$ — the algorithmic complexity of its polarization history along the depth axis. This bound is strictly positive for all programs and grows monotonically with d for non-trivial programs.*

This bound is the computational content of the $\beta = f(d, K_\psi)$ prediction sharpened in Paper 1 Section 7.2. The function f , refined to its full functional dependences in Paper 3 § 6.2, has its physical meaning fixed by Paper 2: K_ψ is the algorithmic complexity of the program running through the bit's causal history, and the GUP correction at the bit's position is set by the irreducibility of that program. Position measurement of a deeply embedded bit requires reading out its program, which by definition cannot be compressed below $K_\psi(d)$.

6.5 The Role of Volume 1, Paper 3

The selection mechanism described in Section 6.3 is the substrate-level prerequisite for learning, hierarchy, and the emergence of subsystems with internal organization. Paper 3 develops the macroscopic consequences: how programs of increasing causal depth come to dominate the polarization landscape over the long flop-time history of a stable cascade instance, how the substrate concentrates computation into coherent regions through the bistability mechanism established there, how nested programs (programs running inside the spatial support of other programs) produce hierarchical structure, and how the first subsystem — the smallest configuration that transforms patterns rather than merely persisting them — emerges from the same axioms. The dissolution rate of non-closed configurations and the effective error-rate decay with coherence depth are both closed in Paper 3 § 2.5 by derivations consistent with the bounds noted above.

VII. The PGE-Forced Wave-Form of Higher-Order Reinterpretation

The substrate's first-order dynamics are wave propagation — the wave equation derived in Paper 1 § VI on the FCC confirmation field is the substrate-level mechanism by which information propagates across the agent manifold. The structurally prior question we address here is: when reinterpretation at any hierarchical level introduces a higher-order dynamical phase, what form does that higher-order dynamics take, and why?

We show that higher-order reinterpretation is *wave-form by structural necessity*, not by additional axiom. The argument is four steps under PGE. The theorem developed here is invoked by Paper 5 § III to resolve the apparent tension between the first-order substrate wave equation and the second-order wave structure that reinterpretation produces.

7.1 The Four-Step Argument

Theorem 0.0.21 (PGE-Forced Wave-Form of Reinterpretation). *The Principle-of-Generative-Economy-selected continuation of the substrate at any reinterpretation boundary takes wave-form dynamics.*

Proof. Four steps.

(i) *First-order substrate behavior is wave propagation.* By Paper 1 § VI, the confirmation field ϕ on the FCC agent manifold satisfies the wave equation $\partial^2 \phi / \partial t^2 - c^2 \nabla^2 \phi = 0$ in the continuum limit, with $c = c_{\text{TMS}} = 1/\sqrt{2}$ lattice spacings per flop. Wave propagation is the substrate's native vocabulary for moving information across the agent manifold.

(ii) *PGE forbids importing mechanisms outside the substrate's vocabulary.* The Principle of Generative Economy, as developed in Paper 1 § 1.3 and applied throughout this volume, selects the minimum sufficient closure at every scale. A continuation that requires a new propagation mechanism — one that does not already operate at the substrate level — adds structure beyond what the existing geometry forces. PGE excludes this. Any higher-order dynamics that PGE selects must reuse the substrate's existing propagation vocabulary.

(iii) *The substrate's only propagation vocabulary is wave propagation.* The FCC geometry, the triadic confirmation rule, and the $[[4,2,2]]$ stabilizer structure collectively determine the substrate's repertoire of information-moving operations. The wave equation is the unique continuum-limit propagation rule that this geometry supports — first established at substrate level in Paper 1, and shown to be form-invariant under recursive coarse-graining (G operator) in § IV of this paper. There is no second native propagation mechanism. The substrate's vocabulary contains exactly one verb for moving information through agent space: the wave.

(iv) *Therefore higher-order reinterpretation is wave-form.* Combining (ii) and (iii): since PGE forbids importing new mechanisms and the only available mechanism is wave propagation, the PGE-selected continuation at a reinterpretation boundary uses wave propagation at the new scale. The recursion is therefore a *wave hierarchy*. Each order of wave reads the structure left by the previous orders, propagating in the level-k flop time at the level-k propagation speed $c^{(k)}$.

□

Status of this result. *Derived.* The four-step argument follows from the wave equation of Paper 1 § VI, the PGE statement of Paper 1 § 1.3, and the form-invariance of the wave equation under G that we established earlier in this paper. No additional structure is introduced.

7.2 Quantization and the Erosion Picture (Previews of Paper 5 § 3.5)

Remark on quantization (preview of Paper 5 § 3.5). A wave hierarchy with closure constraints at each level forces a specific mechanism for stable excitations: packets must settle into configurations where the wave geometry permits standing-wave closure — wave troughs in the accumulated wake of prior propagation. Quantization is not a separate axiom; it is what falls out of having a recursive wave hierarchy at all. This is developed structurally in Paper 5 § 3.5 and applied to field content in Paper 5 § IV.

Remark on the erosion picture. The recursive wave hierarchy is, equivalently, an accumulating wake: every prior propagation event leaves a residue in the substrate's effective topography, and new propagation events ride atop the topography that prior events have already carved. At any given moment, the substrate's working geometry is a superposition of an ancient layer (deep channels carved by very early dynamics, effectively rigid at any subsystem-accessible timescale), an intermediate layer (ongoing dynamics across the current epoch), and a transient layer (flash-flood channels carved by recent events that have not yet relaxed).

This picture is developed for cosmological purposes in Paper 5 § 3.5 and § V; the bare structural claim — that field-effective stability is an erosion property of the wave hierarchy rather than a fundamental axiom — is what the four-step theorem above establishes.

VIII. Predictions and Directions for Subsequent Work

8.1 Empirical Signatures of Polarization Refraction

Although the polarization field ψ propagates at $c_{\text{TMS}} = 1/\sqrt{2}$ across the bulk lattice (Section 2.3) — the same speed as ϕ — its interaction with the confirmation density at the leading edge of a cascade can produce apparent variations in effective propagation velocity. In regions where ϕ is rapidly varying (sharp confirmation gradients) and the polarization landscape is structured, coherent interference between the ϕ and ψ wave fields produces what an external observer would interpret as a position-dependent index of refraction.

This is not a modification of the underlying wave speed. It is interference between two independent fields on the same lattice, observed at coarse-grained scale. The phenomenon is parallel to real optical refraction in standard physics, where the underlying electromagnetic field travels at c everywhere and the apparent slowdown in a dielectric medium is interference with the medium's polarization response. The TMS provides a geometric microstructure for this same phenomenon.

A specific prediction: the apparent variation in effective propagation velocity scales with the local polarization gradient $|\nabla\psi|$ and the local confirmation density ϕ . Regions with steep ψ — gradients (sharp boundaries between coherent polarization domains) act as effective interfaces. Regions with shallow ψ — gradients propagate signals at the bare lattice speed. The functional form is derivable from the coupled wave equations of (ϕ, ψ) and is developed further in Paper 3 alongside the related dissolution-rate and error-decay results.

8.2 The Sharpened GUP Prediction

Paper 1 Section 7.2 introduced the prediction $\beta = f(d, K_\psi)$. Paper 2 has filled in the meaning of K_ψ : it is the Kolmogorov complexity of the program running through the bit's causal history, where “program” has the formal definition of Section 6.2. The prediction therefore acquires its computational content here. Bits embedded in deep, computationally rich programs (high K_ψ at high d) exhibit measurably larger GUP corrections than bits in shallow, computationally trivial configurations at comparable energy scales.

This is a falsifiable prediction with two empirical correlates. First, quantum systems with rich internal computational structure (highly entangled, deeply prepared states) should exhibit measurably larger minimum-uncertainty floors than systems with trivial internal structure at the same total energy. Second, the GUP correction should not be a fixed property of a particle species but should vary with the particle's preparation history. Both are accessible in principle to Planck-scale phenomenology programs (Hossenfelder, 2013) and to next-generation quantum optomechanics experiments. Paper 3 § 6.2 further refines f to account for surround-imprinting and subsystem context, broadening the empirical correlates further.

8.3 The Born Rule from Coarse-Graining

The full quadratic Born rule of standard quantum mechanics is a target for subsequent work. The structural correspondence of Section IV establishes a linear amplitude-to-probability mapping for two-outcome projective measurement with real polarization amplitudes. The recovery of the quadratic $|\cdot|^2$ mapping requires complex amplitudes, which in the TMS are expected to arise from the phase structure of coupled (ψ, Π_ψ)

propagation across many flops. The precise derivation is deferred. The key conceptual claim has already been established: the substrate provides a single mechanism — polarization-biased confirmation against U_{sh} — that unifies what standard quantum mechanics treats as two separate postulates (unitary evolution and measurement collapse).

8.4 Programs, Particles, and Subsequent Volumes

The propagating programs of Section 6.2 — periodic polarization configurations that translate under T^n — are the substrate-level structures from which physical matter is expected to emerge in subsequent volumes. The connection is anticipated rather than asserted here. The substrate-level claim is independently meaningful: the TMS supports a discrete spectrum of stable propagating computational patterns. Whether and how these correspond to the particle spectrum of the Standard Model is the subject of later work in Volume 1 and Volume 2.

Conclusion: A Complete Computational Substrate

Building on the geometric and ontological architecture of Paper 1, Paper 2 has shown that the TMS substrate supports computation by geometric necessity, with no additional axioms or free parameters. The following results emerge from the primitives of Paper 1 alone:

- The polarization field ψ propagates by a homogeneous wave equation on the FCC lattice at $c_{\text{TMS}} = 1/\sqrt{2}$, identical in form to the confirmation field ϕ — the propagation speed and lattice structure are inherited, not re-derived.
- The conjugate pair (ψ, Π_ψ) emerges from the same conservation argument that produced (ϕ, Π) in Paper 1, applied to polarization content rather than confirmation count.
- Synthetic bits are geometric appointments rather than valued outcomes — a distinction forced by Axiom 3 (no value can be inherited without a rule; no value can be independent without dissipating polarization signals) and the Principle of Generative Economy.
- Confirmation events resolve synthetic bit positions into binary values via a linear polarization-to-probability mapping $P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi)/2$ — the unique mapping consistent with signal conservation, the maximum-entropy character of U_{sh} , and Generative Economy.
- This mapping constitutes a structural correspondence to the Born rule for two-outcome projective measurement, paralleling the $[[4,2,2]]$ structural correspondence of Paper 1 in status.
- Each confirmation event is a probabilistic 3-input majority gate; the $[[4,2,2]]$ stabilizer repair mechanism projects local errors toward stabilizer-satisfying configurations; the effective behavior in coherent regions is deterministic majority computation.
- The substrate supports universal classical computation by the universality of $\{\text{MAJ}, \text{constant}\}$.
- Programs — configurations causally closed under the flop operator with $K(C) < |C|$ — are selected automatically by the stabilizer repair mechanism. Configurations that compute persist; configurations that do not compute dissolve into U_{sh} — like noise.
- The $\beta = f(d, K_\psi)$ prediction from Paper 1 acquires its computational meaning: K_ψ is the Kolmogorov complexity of the program running through a bit's causal history, and the GUP correction at the bit's position is the irreducibility of that program made geometrically visible.

The arc from Paper 1 to Paper 2 is a single unbroken chain: U_{sh} provides the maximum-entropy substrate; triadic confirmation pulls binary outcomes from it; the FCC lattice carries those outcomes forward as wave

dynamics; the polarization field encodes their computational content; the Born-rule structural correspondence governs their per-event resolution; the majority gate is the irreducible computational primitive; the stabilizer dynamics select programs from noise. No new axioms. No new parameters. No new mechanisms.

The remaining derivations of Volume 1 — learning and hierarchy (Paper 3), subsystem emergence (Paper 4), and simulation (Paper 5) — proceed from this computational substrate. The agent-side complement of the framework, addressing the relationship between substrate-level computation and observer experience, is the subject of Volume 2.

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